

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397361
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Zaffetti; Mark A. et al.

Control assembly fabrication via brazing

Abstract

Systems and methods to control fabrication of an assembly involve a first end sheet having an interior surface and an exterior surface, opposite the interior surface. A system includes interior sheets, the interior sheets including a first interior sheet at one end of a stack of the interior sheets and including a last interior sheet at an opposite end of the stack of the two or more interior sheets, the first interior sheet being adjacent to the interior surface of the first end sheet, and a second end sheet having an interior surface and an exterior surface, opposite the interior surface. The last interior sheet is adjacent to the interior surface of the last end sheet, and the interior surface of the first end sheet or the second end sheet includes venting features, the venting features including holes or slots to channel heat, gas, or vapor during a brazing process.

Inventors: Zaffetti; Mark A. (Suffield, CT), Abraham; Benjamin (Marlborough, CT), Sauerhoefer; Timothy (Broad Brook, CT)

Applicant: Hamilton Sundstrand Corporation (Charlotte, NC)

Family ID: 1000008777943

Assignee: HAMILTON SUNDSTRAND CORPORATION (Charlotte, NC)

Appl. No.: 18/192777

Filed: March 30, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230249273 A1	Aug. 10, 2023

Related U.S. Application Data

division parent-doc US 17060359 20201001 US 11633799 child-doc US 18192777

Publication Classification

Int. Cl.: B23K1/00 (20060101); B23K1/018 (20060101); B23K101/14 (20060101)

U.S. Cl.:

CPC B23K1/018 (20130101); B23K1/0012 (20130101); B23K2101/14 (20180801)

Field of Classification Search

CPC: F28D (9/0043); F28D (9/005); B23K (1/0012); B23K (26/342); B23K (35/0233); B23K (2101/14)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3375570	12/1967	Dubusker	228/183	B23K 1/0012
4247508	12/1980	Housholder	264/122	B29C 64/124
5931224	12/1998	Chevallier	165/DIG.466	F28F 17/005
6216775	12/2000	Hedman	165/96	F28F 13/12
6408940	12/2001	Naji	165/906	F28F 9/0214
6554600	12/2002	Hofmann	425/182	B29C 41/34
6843311	12/2004	Evans	165/167	F28D 9/005
7204684	12/2006	Ederer	425/375	B29C 64/259
7533717	12/2008	Hummel	165/178	F28F 9/0075
7753105	12/2009	Acre	165/41	B60H 1/3227
8741194	12/2013	Ederer	264/236	B22C 1/162
9079248	12/2014	Ackelid	N/A	B22F 3/105
9163882	12/2014	Crawford	N/A	N/A
9597730	12/2016	Mironets	N/A	B23K 26/083
9614206	12/2016	Choi et al.	N/A	N/A
9638471	12/2016	Neumann	N/A	F28D 15/00
9914169	12/2017	Ederer	N/A	B22F 10/14
10006722	12/2017	Kenney et al.	N/A	N/A
10302366	12/2018	Schatz-Knecht	N/A	F28F 3/08
10413968	12/2018	Pialot	N/A	B29C 64/245
10801789	12/2019	Subramanyam et al.	N/A	N/A
10907906	12/2020	Sun	N/A	F28F 3/046
10935330	12/2020	Subramanyam et al.	N/A	N/A
11274884	12/2021	Thomas	N/A	F28D 9/005
2001/0045678	12/2000	Kubo	425/375	B29C 64/35
2002/0066552	12/2001	Komoda	165/134.1	F28F 19/004
2003/0004599	12/2002	Herbak	700/98	B29C 67/246
2003/0083771	12/2002	Schmidt	700/118	B33Y 10/00
2004/0025905	12/2003	Ederer	15/300.1	B22C 23/00
2004/0026418	12/2003	Ederer	220/23.86	B29C 64/259

2005/0155749	12/2004	Memory	165/167	F28D 9/005
2005/0186538	12/2004	Uckelmann	433/201.1	B33Y 80/00
2005/0194123	12/2004	Strahle	165/167	F28F 3/048
2007/0026099	12/2006	Hagiwara	425/174.4	B29C 64/153
2007/0057412	12/2006	Weiskopf	425/375	B33Y 30/00
2007/0087071	12/2006	Devos	425/375	B29C 64/165
2007/0298182	12/2006	Perret	118/308	B22F 12/13
2008/0190905	12/2007	Heinlein	219/121.73	B29C 64/153
2008/0233302	12/2007	Elsner	427/421.1	B33Y 70/00
2009/0152771	12/2008	Philippi	264/410	B33Y 30/00
2010/0007062	12/2009	Larsson	264/485	B22F 10/362
2010/0212584	12/2009	Ederer	118/56	B29C 64/165
2010/0291314	12/2009	Kashani-Shirazi	524/502	B29C 64/153
2010/0323301	12/2009	Tang	430/325	G03F 7/0037
2011/0059247	12/2010	Kuzusako	427/372.2	B22C 7/02
2011/0133367	12/2010	Weidinger	264/497	B33Y 40/00
2011/0293771	12/2010	Oberhofer	425/182	B29C 64/364
2012/0119399	12/2011	Fruth	425/186	B29C 64/245
2012/0170222	12/2011	Dede et al.	N/A	N/A
2012/0267084	12/2011	Crawford	165/170	F28F 3/005
2013/0026680	12/2012	Ederer	425/166	B22F 12/55
2013/0064707	12/2012	Matsui	264/109	B29C 64/176
2014/0048980	12/2013	Crump	264/401	B29C 64/106
2014/0265049	12/2013	Burris	264/497	B22F 12/90
2014/0271961	12/2013	Khoshnevis	425/375	B28B 1/001
2014/0306379	12/2013	Hartmann	425/130	B29C 64/118
2014/0370323	12/2013	Ackelid	428/548	B23K 15/00
2015/0042018	12/2014	Gunther	425/150	B22F 12/22
2015/0054204	12/2014	Tseliakhovich	425/162	B29C 64/264
2015/0273572	12/2014	Ederer	164/15	B29C 64/165
2016/0204486	12/2015	Kenney et al.	N/A	N/A
2016/0303762	12/2015	Günther	N/A	B28B 1/001
2016/0315365	12/2015	Vanderwees et al.	N/A	N/A
2017/0038150	12/2016	Kalbacher et al.	N/A	N/A
2017/0106603	12/2016	Pobihun	N/A	B29C 64/135
2017/0120332	12/2016	DeMuth	N/A	B29C 64/264
2017/0241716	12/2016	Schatz-Knecht	N/A	N/A
2017/0267991	12/2016	Lipkens	N/A	C12M 33/10
2017/0334134	12/2016	Herzog	N/A	B29C 64/182
2017/0361525	12/2016	Warner	N/A	B33Y 10/00
2018/0035689	12/2017	Warner	N/A	B29C 48/92
2018/0236504	12/2017	Pourcher	N/A	B08B 5/04
2018/0263081	12/2017	Dardona	N/A	B33Y 80/00
2018/0264726	12/2017	Shiomi	N/A	B29C 64/268
2018/0281295	12/2017	Tibbits	N/A	B29C 64/106
2018/0304353	12/2017	LaPlante	N/A	B22F 12/222
2018/0304369	12/2017	Myerberg	N/A	B22F 10/18
2019/0017748	12/2018	Sun	N/A	F28F 3/048
2019/0076924	12/2018	Jepeal	N/A	B33Y 70/00
2019/0134908	12/2018	Chanclon	N/A	B33Y 50/02

2019/0151973	12/2018	Itoh	N/A	B23K 35/286
2019/0168305	12/2018	Boyle	N/A	B29C 64/386
2019/0210106	12/2018	Gibson	N/A	B22F 3/1025
2019/0257595	12/2018	Barriga et al.	N/A	N/A
2019/0323778	12/2018	Ojira	N/A	F24H 1/12
2020/0049413	12/2019	Ojira	N/A	F28D 9/0056
2020/0086568	12/2019	Wynne	N/A	B29C 64/124
2020/0139411	12/2019	Zaveri	N/A	B08B 3/12
2020/0191436	12/2019	Ono	N/A	F24H 8/00
2020/0198238	12/2019	Eideloth	N/A	B01D 67/00415
2020/0324479	12/2019	Friedrich	N/A	B08B 3/10
2021/0032034	12/2020	Kalouche	N/A	B25J 9/1612
2022/0105582	12/2021	Zaffetti et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102015010310	12/2016	DE	N/A
3204183	12/2016	EP	N/A

OTHER PUBLICATIONS

Extended European Search Report for EP Application No. 21186462.4, dated Jan. 19, 2022, pp. 1-7. cited by applicant

EPO Official Letter for Application No. 21186462.4, mailed Nov. 22, 2023, 5 pages. cited by applicant

Primary Examiner: Saad; Erin B

Attorney, Agent or Firm: CANTOR COLBURN LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a division of U.S. application Ser. No. 17/060,359 filed Oct. 1, 2020, the entire contents of which are incorporated herein by reference.

BACKGROUND

(1) Exemplary embodiments pertain to the art of assembly fabrication and, in particular, to control of assembly fabrication via brazing.

(2) Brazing is a process by which two or more metal components are joined together by melting and flowing a filler metal into the interface. The filler metal has a lower melting point than the metal components being joined. Exemplary filler materials include aluminum-silicon, copper, copper-silver, copper-zinc (i.e., brass), copper-tin (i.e., bronze), and nickel alloy.

BRIEF DESCRIPTION

(3) In one embodiment, a system to control fabrication of an assembly includes a first end sheet having an interior surface and an exterior surface, opposite the interior surface. The system also includes two or more interior sheets, the two or more interior sheets including a first interior sheet at one end of a stack of the two or more interior sheets and including a last interior sheet at an opposite end of the stack of the two or more interior sheets, the first interior sheet being adjacent to

the interior surface of the first end sheet. A second end sheet has an interior surface and an exterior surface, opposite the interior surface. The last interior sheet of the two or more interior sheets is adjacent to the interior surface of the last end sheet. The interior surface of the first end sheet or the interior surface of the second end sheet includes one or more venting features, the one or more venting features including holes or slots to channel heat, gas, or vapor during a brazing process to braze the first end sheet, the two or more interior sheets, and the second end sheet into the assembly.

(4) Additionally or alternatively, in this or other embodiments, the one or more venting features include a series of holes that channel the heat, the gas, or the vapor from the interior surface through to the exterior surface of the first end sheet or the second end sheet.

(5) Additionally or alternatively, in this or other embodiments, the series of holes have a geometric shape.

(6) Additionally or alternatively, in this or other embodiments, different holes in the series of holes have different shapes.

(7) Additionally or alternatively, in this or other embodiments, the one or more venting features include slots that channel the heat, the gas, or the vapor from a central portion of the interior surface to edges of the interior surface of the first end sheet or the second end sheet.

(8) Additionally or alternatively, in this or other embodiments, the slots are formed in a pattern.

(9) Additionally or alternatively, in this or other embodiments, the pattern is a checkerboard or the pattern includes concentric circles with diagonal slots that cross at a center of the concentric circles and extend to the edges of the interior surface of the first end sheet or the second end sheet.

(10) Additionally or alternatively, in this or other embodiments, the one or more venting features include slots that channel the heat, the gas, or the vapor to a particular area of the interior surface to cause incomplete brazing at the particular area between the interior surface of the first end sheet and the first interior sheet or between the interior surface of the second end sheet and the last interior sheet.

(11) Additionally or alternatively, in this or other embodiments, the venting features include both holes and slots.

(12) Additionally or alternatively, in this or other embodiments, both the interior surface of the first end sheet and the interior surface of the second end sheet include venting features.

(13) In another embodiment, a method of controlling a brazing process to fabricate an assembly includes configuring a stack of layers. The stack includes a first end sheet having an interior surface and an exterior surface, opposite the interior surface. The stack also includes two or more interior sheets, the two or more interior sheets including a first interior sheet at one end of a stack of the two or more interior sheets and including a last interior sheet at an opposite end of the stack of the two or more interior sheets, the first interior sheet being adjacent to the interior surface of the first end sheet. A second end sheet of the two or more interior sheets has an interior surface and an exterior surface, opposite the interior surface, wherein the last interior sheet of the two or more interior sheets is adjacent to the interior surface of the last end sheet. The method includes including one or more vent features in the interior surface of the first end sheet or the interior surface of the second end sheet, the one or more venting features including holes or slots to channel heat, gas, or vapor during the brazing process used to braze the first end sheet, the two or more interior sheets, and the second end sheet to fabricate the assembly.

(14) Additionally or alternatively, in this or other embodiments, the one or more venting features include a series of holes that channel the heat, the gas, or the vapor from the interior surface through to the exterior surface of the first end sheet or the second end sheet.

(15) Additionally or alternatively, in this or other embodiments, the series of holes have a geometric shape.

(16) Additionally or alternatively, in this or other embodiments, different holes in the series of holes have different shapes.

- (17) Additionally or alternatively, in this or other embodiments, the one or more venting features include slots that channel the heat, the gas, or the vapor from a central portion of the interior surface to edges of the interior surface of the first end sheet or the second end sheet.
- (18) Additionally or alternatively, in this or other embodiments, the slots are formed in a pattern.
- (19) Additionally or alternatively, in this or other embodiments, the pattern is a checkerboard or the pattern includes concentric circles with diagonal slots that cross at a center of the concentric circles and extend to the edges of the interior surface of the first end sheet or the second end sheet.
- (20) Additionally or alternatively, in this or other embodiments, the one or more venting features include slots that channel the heat, the gas, or the vapor to a particular area of the interior surface to cause incomplete brazing at the particular area between the interior surface of the first end sheet and the first interior sheet or between the interior surface of the second end sheet and the last interior sheet.
- (21) Additionally or alternatively, in this or other embodiments, the venting features include both holes and slots.
- (22) Additionally or alternatively, in this or other embodiments, the including the venting features includes including the venting features on both the interior surface of the first end sheet and the interior surface of the second end sheet.
-

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
- (2) FIG. 1 is an exploded view of aspects of an exemplary assembly that may be fabricated according to one or more embodiments;
- (3) FIG. 2 shows a fabricated exemplary assembly according to one or more embodiments;
- (4) FIG. 3 shows exemplary venting features that facilitate control of assembly fabrication via brazing according to one or more embodiments;
- (5) FIG. 4 shows an exemplary end sheet of an assembly that facilitates control of assembly fabrication via brazing based on slots according to one or more embodiments;
- (6) FIG. 5 is a cross-sectional view through A-A indicated in FIG. 4;
- (7) FIG. 6 shows an exemplary end sheet of an assembly that facilitates control of assembly fabrication via brazing based on an exemplary pattern of slots according to one or more embodiments;
- (8) FIG. 7 shows an interior surface of an exemplary end sheet that facilitates control of assembly fabrication via brazing according to one or more embodiments;
- (9) FIG. 8 shows an exemplary end sheet of an assembly that facilitates control of assembly fabrication via brazing based on holes according to one or more embodiments; and
- (10) FIG. 9 shows an exemplary end sheet of an assembly that facilitates control of assembly fabrication via brazing based on an exemplary pattern of holes according to one or more embodiments.

DETAILED DESCRIPTION

- (11) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.
- (12) As previously noted, brazing is a process by which metal components are joined. The filler material used in brazing generally has a higher melting point than the filler used in soldering and, thus, produces a stronger joint. Brazing is widely used in various applications. For example, brazing preforms may be used to attach electronic components with silicon dies to a substrate (e.g., printed circuit board). As another example, brazed plate heat exchangers include plates (e.g.,

copper, stainless steel) that are stacked and brazed together to aggregate the heat transfer area over multiple layers of the stack in a compact assembly. The brazing rather than bolting, for example, means that rubber gaskets are not needed between adjacent plates. Yet another example, used herein for explanatory purposes is a brazed plate and fin heat exchanger that includes plates and fins stacked as interior components between end sheets and brazed together.

(13) Embodiments of the systems and methods detailed herein relate to control of assembly fabrication via brazing. An assembly (e.g., plate and fin heat exchanger) includes two end sheets or outer sheets on each end of the assembly. The outer sheets sandwich the other components that make up the assembly. In prior assemblies formed via brazing (i.e., brazed assemblies) with solid end sheets, heat, gas, and vapor trapped between an outer sheet and the next sheet of the assembly during brazing may result in incomplete brazing. The heat, gas, and vapor that prevent complete brazing in some areas result in a weaker bond in those areas. When the brazing process requires a fast cooling rate from the brazing temperature in order to achieve specific material properties, partially brazed areas can also result. As the footprint of the assembly grows, the distance that heat, gas, and vapor must travel to vent out of the assembly also necessarily grows. Thus, issues with the prior brazing process may be exacerbated.

(14) According to one or more embodiments, control of assembly fabrication via brazing is achieved by incorporating venting features into the outer sheets of the assembly, as detailed. Exemplary venting features include holes, slots, or a combination of the two. One exemplary form of control involves venting heat, gas, and vapor out of the assembly via the venting features to achieve complete and uniform brazing. Another exemplary form of control involves directing heat, gas, and vapor to specific areas to achieve an incomplete brazing only where desired.

(15) FIG. 1 is an exploded view of aspects of an exemplary assembly **100** that may be fabricated via brazing according to one or more embodiments. The exemplary assembly **100** is a plate and fin heat exchanger. The exploded view in FIG. 1 shows the layers that are brazed to form the assembly **100** shown in FIG. 2. Two end sheets **110a**, **110b** (generally referred to as **110**) sandwich the interior components **130**. Each of the end sheets **110** has an interior surface **115a**, **115b** (generally referred to as **115**) that is adjacent to the interior components **130**. In the view shown in FIG. 1, the interior sheet **115a** of the end sheet **110a** is not visible but the interior surface **115b** of the end sheet **110b** is visible. Each of the end sheets **110** also has an exterior surface **120a**, **120b** (generally referred to as **120**) that is opposite the corresponding interior surface **115a**, **115b**. In the view shown in FIG. 1, the exterior surface **120a** of the end sheet **110a** is visible but the exterior surface **120b** of the end sheet **110b** is not visible.

(16) The interior components **130** that are arranged in a stack, as shown, include a first interior sheet **140** that is adjacent to the end sheet **110a** and, specifically, to the interior surface **115a** of the end sheet **110a**. The interior components **130** also include a last interior sheet **150** that is adjacent to the end sheet **110b** and, specifically, to the interior surface **115b** of the end sheet **110b**. More particularly, a side of the first interior sheet **140** that is opposite the side that is adjacent to the other interior components **130** is adjacent to the interior surface **115a** of the end sheet **110a** and is brazed to the interior surface **115a** of the end sheet **110a**. And, a side of the last interior sheet **150** that is opposite the side that is adjacent to the other interior components **130** is adjacent to the interior surface **115b** of the end sheet **110b** and is brazed to the interior surface **115b** of the end sheet **110b**.

(17) The exemplary end sheets **110** and interior components **130** of the exemplary assembly **100** pertain to a heat exchanger. For example, the interior components **130** include interleaved sets of a hot fin **131** and hot closure bars **132** and a cold fin **135** and cold closure bars **136** with parting sheets **145** that separate the interleaved sets. The first interior sheet **140** and the last interior sheet **150** in the exemplary assembly **100** are parting sheets **145**, as shown. The dimensions (i.e., length, width, thickness) of the components (end sheets **110**, interior components **130**) are not intended to be limited by the example shown in FIG. 1. The material of the end sheets **110** and interior components **130** is based on the type of assembly **100**. For example, an aluminum heat exchanger

may include end sheets **110** of 3003 aluminum or 3004 aluminum, fins of 6951 aluminum, and parting sheets of a 6000 series aluminum core that is clad on both sides with braze alloy. As another example, a steel heat exchanger may include components that are mostly 347 stainless steel with some fins of nickel 201 and others of 347 stainless steel.

(18) As discussed with reference to FIG. 3, the end sheets **110** include venting features **300** to facilitate control during the brazing process to fabricate the assembly **100** according to one or more embodiments. According to an exemplary embodiment, the venting features **300** may ensure that heat, gas, and vapor produced during the brazing process to fabricate the assembly **100** from the end sheets **110** and the interior components **130** are vented out. This ensures uniform and complete brazing between the interior surface **115a** of the end sheet **110a** and the first interior sheet **140** or between the interior surface **115b** of the end sheet **110b** and the last interior sheet **150**. According to other exemplary embodiments, the venting features **300** may channel heat, gas, and vapor produced during the brazing process to particular areas **710** (FIG. 7) of the interface between the interior surface **115a** of the end sheet **110a** and the first interior sheet **140** or of the interface between the interior surface **115b** of the end sheet **110b** and the last interior sheet **150** in order to control incomplete brazing in those particular areas **710**.

(19) FIG. 2 shows a fabricated exemplary assembly **100** according to one or more embodiments. The end sheets **110** and the interior components **130** are brazed together, as indicated. The interior sheets **130** that are sandwiched between the two end sheets **110** and one of the end sheets **110** are not visible in the view shown in FIG. 2. The exemplary assembly **100** is a plate and fin heat exchanger. As such, an inlet **220a** and outlet **230a** for a first medium and an inlet **220b** and outlet **230b** for a second medium are shown. One of the media (i.e., the first medium or the second medium) transfers heat to the other while traversing separate paths through the assembly **100**.

(20) FIG. 3 shows exemplary venting features **300** that facilitate control of assembly fabrication via brazing according to one or more embodiments. A different type of venting feature **300** is shown on each end of the end sheet **110** in FIG. 3. As one of the magnified views indicates, the type of venting feature **300** shown on one end of the end sheet **110** is holes **310**. These holes **310** go through the end sheet **110** and channel heat, gas, and vapor out of the assembly **100** during the brazing process. The exemplary holes **310** shown in FIG. 3 are not intended to limit the numbers and positions of the holes **310** that may be incorporated in one or more end sheets **110** in order to vent heat, gas, and vapor out of the assembly **100** during brazing of the end sheet **100** to either the first interior sheet **140** or the last interior sheet **150**. Additional examples are discussed with reference to FIGS. 8 and 9.

(21) As the other magnified view indicates, the type of venting feature **300** shown on another end of the end sheet **110** is slots **320**. Unlike the holes **310**, the slots **320** are formed as a trench in the interior surface **115**. As shown, these slots **320** channel heat, gas, and vapor to the edges of the end sheet **110** and out of the assembly **100**. However, as discussed with reference to FIG. 7, one or more slots **320** may be used to channel heat, gas, and/or vapor to a particular area **710** of the interface between the end sheet **110** and the first interior sheet **140** or the last interior sheet **150** during brazing to controllably create an area **710** of incomplete brazing. While holes **310** and slots **320** may both be formed on a given end sheet **110**, as shown in FIG. 3, only one type of venting feature **300** may be included, as well, as shown in FIGS. 4-9.

(22) FIG. 4 shows an exemplary end sheet **110** used to control assembly fabrication via brazing according to one or more embodiments. The interior surface **115** of the end sheet **110** is visible in the view of FIG. 4. The venting features **300** included on the interior surface **115** are slots **320** formed in a checker pattern. Each of the slots **320** extends from one edge to an opposite edge. As such, heat, gas, and vapor generated during brazing of the end sheet **110** to a first interior sheet **140** or a last interior sheet **150** may be channeled out at the edges via the slots **320**, as indicated by the arrows. The cross-section A-A indicated in FIG. 4 is shown in FIG. 5.

(23) FIG. 5 is a cross-sectional view A-A indicated in FIG. 4. The slots **320** extending in one

direction are shown by the cross-sectional view. As previously noted, the slots **320** are formed as trenches in the interior surface **115** of the end sheet **110**. The shape and depth of the slots **320** is not intended to be limited by the example shown. The shape and depth of the slots **320** may be based on the overall dimensions of the end sheet **110**. Further, different slots **320** on the same interior surface **115** may have different shapes and depths. The selection of the location and characteristics of the slots **320** may be based on thermal analysis and other studies regarding the brazing process. These studies may be particular to the materials of the end sheet **110** and the first interior sheet **140** or last interior sheet **150**, as well as to their dimensions.

(24) FIG. **6** shows an exemplary end sheet **110** that facilitates control of assembly fabrication via brazing according to one or more embodiments. The interior surface **115** of the end sheet **110** is visible in the view of FIG. **6**. The venting features **300** included on the interior surface **115** are slots **320** formed in concentric circles and diagonal lines. As shown, the slots **320** formed as concentric circles channel heat, gas, and vapor to the diagonal lines, which vent the heat, gas, and vapor toward the edges and out at the corners of the end sheet **110** in the exemplary case.

(25) FIG. **7** shows an interior surface **115** of an exemplary end sheet **110** that facilitates control of assembly fabrication via brazing according to one or more embodiments. As previously noted, control during brazing is not limited to venting out heat, gas, and vapor to facilitate uniform and complete brazing between the end sheet **110** and either the first interior sheet **140** or the last interior sheet **150**. The exemplary slot **320** shown in FIG. **7** is a venting feature **300** that controllably creates an area **710** of incomplete brazing.

(26) That is, when the first interior sheet **140** or the last interior sheet **150** is brazed to the interior surface **115** shown in FIG. **7**, some of the heat, gas, and vapor generated during the brazing process will be channeled to the area **710**. This channeling path is indicated by the arrow along the slot **320**. As a result of the heat, gas, and/or vapor in the area **710**, the end sheet **110** and the first interior sheet **140** or the last interior sheet **150** will not be completely brazed in the area **710** in the resulting assembly **100**. An end sheet **110** may include slots **320** and holes **310** that vent heat, gas, and vapor as well as slots **320** that channel heat, gas, and vapor to one or more areas **710** of the interior surface **115**.

(27) FIG. **8** shows an exemplary end sheet **110** of an assembly **100** that facilitates control of assembly fabrication via brazing according to one or more embodiments. The interior surface **115** of the end sheet **110** is visible in the view of FIG. **8**. The venting features **300** included on the interior surface **115** are holes **310**. The exemplary holes **310** shown in FIG. **8** illustrate that the holes **310** may be of different shapes and sizes and may, as shown in FIG. **8**, be a mix of different types of holes **310**. The holes **310** may have geometric shapes (e.g., rectangle, oval, diamond) or irregular shapes. As with the slots **320**, the locations and characteristics of the holes **310** may be selected based on an analysis of the brazing process for the specific dimensions and materials of the end sheets **110** and first interior sheet **140** or last interior sheet **150** involved.

(28) FIG. **9** shows an exemplary end sheet **110** of an assembly **100** that facilitates control of assembly fabrication via brazing according to one or more embodiments. The interior surface **115** of the end sheet **110** is visible in the view of FIG. **9**. The venting features **300** included on the interior surface **115** are holes **310** arranged in a pattern with a concentration of the holes **310** in the center of the interior surface **115** where heat, gas, and vapor may be concentrated during the brazing process. This exemplary pattern does not limit the patterns, sizes, and shapes of holes **310** that may be used. As previously noted, slots **320** may be used in conjunction with the holes **310**, as well.

(29) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements,

and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(30) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A method of controlling a brazing process to fabricate an assembly, the method comprising: configuring a stack of layers, the stack comprising: a first end sheet having an interior surface and an exterior surface, opposite the interior surface, two or more interior sheets, the two or more interior sheets including a first interior sheet at one end of a stack of the two or more interior sheets and including a last interior sheet at an opposite end of the stack of the two or more interior sheets, the first interior sheet being adjacent to the interior surface of the first end sheet, a second end sheet having an interior surface and an exterior surface, opposite the interior surface, wherein the last interior sheet of the two or more interior sheets is adjacent to the interior surface of the last end sheet; and including one or more vent features in the interior surface of the first end sheet or the interior surface of the second end sheet, the one or more venting features including holes or slots configured to channel heat, gas, or vapor during the brazing process used to braze the first end sheet, the two or more interior sheets, and the second end sheet to fabricate the assembly; wherein the one or more venting features include slots that channel the heat, the gas, or the vapor to a particular area of the interior surface to cause incomplete brazing at a particular area between the interior surface of the first end sheet and the first interior sheet or between the interior surface of the second end sheet and the last interior sheet.

2. A method of controlling a brazing process to fabricate an assembly, the method comprising: configuring a stack of layers, the stack comprising: a first end sheet having an interior surface and an exterior surface, opposite the interior surface, two or more interior sheets, the two or more interior sheets including a first interior sheet at one end of a stack of the two or more interior sheets and including a last interior sheet at an opposite end of the stack of the two or more interior sheets, the first interior sheet being adjacent to the interior surface of the first end sheet, a second end sheet having an interior surface and an exterior surface, opposite the interior surface, wherein the last interior sheet of the two or more interior sheets is adjacent to the interior surface of the last end sheet; and including one or more vent features in the interior surface of the first end sheet or the interior surface of the second end sheet, the one or more venting features including holes or slots configured to channel heat, gas, or vapor during the brazing process used to braze the first end sheet, the two or more interior sheets, and the second end sheet to fabricate the assembly; wherein the one or more venting features include slots that channel the heat, the gas, or the vapor from a central portion of the interior surface to edges of the interior surface of the first end sheet or the second end sheet; the slots are formed in a pattern that includes concentric circles with diagonal slots that cross at a center of the concentric circles and extend to the edges of the interior surface of the first end sheet or the second end sheet.

3. The method according to claim 1, wherein including the venting features includes including the venting features on both the interior surface of the first end sheet and the interior surface of the second end sheet.

4. The method according to claim 2, wherein including the venting features includes including the

venting features on both the interior surface of the first end sheet and the interior surface of the second end sheet.
